

Hints & Solutions

#MathBoleTohMathonGo

Q1 (4) - Single Correct

 $a_1, a_2, a_3 \dots a_n$, are in an AP.

$$a_1 + a_3 + a_5 = -12 \text{ and } a_1 a_2 a_3 = 8$$

$$\Rightarrow a_3 = -4 \Rightarrow a_1 a_2 = -2$$

$$a_1 + 2d = -4 \dots (i)$$

$$\Rightarrow a_1(a_1 + d) = -2 \dots (ii)$$

$$d = \left(\frac{-4 - a_1}{2} \right)$$

$$\Rightarrow a_1^2 + a_1 \left(\frac{-4 - a_1}{2} \right) = -2$$

$$\Rightarrow a_1^2 - 4a_1 + 4 = 2$$

$$\Rightarrow a_1 = 2 \text{ and } d = -3$$

Now, value of $a_2 + a_4 + a_6 = 3a_4$

$$= 3(a_1 + 3d)$$

$$= 3[(2 + 3(-3))] = -21$$

Q2 (3) - Single Correct

Given, $S_1 + S_2 + \dots + S_{10} = t_1 + t_2 + \dots + t_{15}$ Let 1st sequence is $a_1, a_1 + d_1, a_1 + 2d_1, \dots$ and 2nd sequence is $a_1, a_1 + d_2, a_1 + 2d_2, \dots$ [$\because S_1 = t_1$]Given, $S_2 = 2t_2$

$$\therefore a_1 + d_1 = 2(a_1 + d_2)$$

$$\Rightarrow a_1 = d_1 - 2d_2 \dots (i)$$

We have to find $\frac{S_2 - S_1}{t_2 - t_1} = \frac{d_1}{d_2} = ?$

$$\text{Now, } \frac{10}{2}[2a_1 + 9d_1] = \frac{15}{2}[2a_1 + 14d_2]$$

$$\Rightarrow 2a_1 + 9d_1 = 3a_1 + 21d_2$$

$$\Rightarrow a_1 = 9d_1 - 21d_2 \dots (ii)$$

From Eqs. (i) and (ii), we get

$$d_1 - 2d_2 = 9d_1 - 21d_2$$

$$\Rightarrow 19d_2 = 8d_1$$

$$\text{or } \frac{d_1}{d_2} = \frac{19}{8}$$

Q3 (3) - Single Correct

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Here,

$$T_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n(2n+2)} \{(2n+2) - (2n+1)\}$$

$$= \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots (2n)} - \frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots (2n)(2n+2)}$$

$$\therefore T_1 = \left(\frac{1}{2} - \frac{1 \cdot 3}{2 \cdot 4} \right)$$

$$T_2 = \left(\frac{1 \cdot 3}{2 \cdot 4} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \right)$$

$$T_3 = \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} - \frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8} \right)$$

$\vdots$

$$T_n = \left[\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots (2n)} - \frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots (2n)(2n+2)} \right]$$

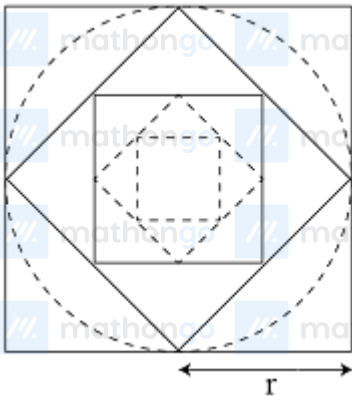
$$\Rightarrow S_n = \frac{1}{2} - \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots (2n)(2n+2)}$$

Note, that $S_\infty = \frac{1}{2}$

Q4 (1) - Single Correct

Radius of the circle is given r . Hence, side of the square is $2r$. Square formed by joining the mid-points of the sides of square will be side of $r\sqrt{2}$.

Hence, radius of the sequence will be $r, \frac{r\sqrt{2}}{2}, \dots$ terms



$$a_n = ar^{n-1}$$

$$a_n = r \left(\frac{1}{\sqrt{2}} \right)^{n-1}$$

$$a_n = r \left(\frac{1}{2} \right)^{\frac{n-1}{2}}$$

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$$a_n = r(2)^{\frac{1-n}{2}}$$

So, radius of n th term is $r \left(2^{\frac{1-n}{2}} \right)$.

Q5 (4) - Single Correct

$$\sum_{k=1}^{360} \left(\frac{1}{k\sqrt{k+1}(k+1)\sqrt{k}} \right) = \sum_{k=1}^{360} \left(\frac{k\sqrt{k+1} - (k+1)\sqrt{k}}{k^2(k+1) - (k+1)^2k} \right)$$

$$\sum_{k=1}^{360} \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right) = 1 - \frac{1}{\sqrt{361}} = \frac{18}{19} = \frac{m}{n}$$

Then, $m + n = 18 + 19 = 37$

Q6 (3) - Single Correct

$$\sum_{n=1}^{\infty} \frac{n}{n^4+4} = \sum_{n=1}^{\infty} \frac{n}{(n^2+2)^2-4n^2}$$

$$= \sum_{n=1}^{\infty} \frac{n}{(n^2+2+2n)(n^2+2-2n)}$$

$$= \sum_{n=1}^{\infty} \frac{1}{4} \left[\frac{1}{n^2+2-2n} - \frac{1}{n^2+2+2n} \right]$$

$$T_n = \frac{1}{4} \left[\frac{1}{n^2+2-2n} - \frac{1}{n^2+2+2n} \right]$$

$$\text{Now, } T_1 = \frac{1}{4} \left[\frac{1}{1} - \frac{1}{5} \right]$$

$$T_2 = \frac{1}{4} \left[\frac{1}{2} - \frac{1}{10} \right]$$

$$T_3 = \frac{1}{4} \left[\frac{1}{5} - \frac{1}{17} \right]$$

$$T_4 = \frac{1}{4} \left[\frac{1}{10} - \frac{1}{26} \right]$$

$$T_5 = \frac{1}{4} \left[\frac{1}{17} - \frac{1}{37} \right]$$

$$\vdots$$

$$T_{n-1} = \frac{1}{4} \left[\frac{1}{(n-1)^2+2-2(n-1)} - \frac{1}{(n-1)^2+2+2(n-1)} \right]$$

$$T_n = \frac{1}{4} \left[\frac{1}{n^2+2-2n} - \frac{1}{n^2+2+2n} \right]$$

On adding all the above terms, we get

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$$S = \frac{1}{4} \left[1 + \frac{1}{2} - \frac{1}{(n-1)^2 + 2 + 2(n-1)} - \frac{1}{n^2 + 2 + 2n} \right]$$

$$S_{\infty} = \lim_{n \rightarrow \infty} \frac{1}{4} \left[\frac{3}{2} - \frac{1}{n^2 + 1} - \frac{1}{n^2 + 2 + 2n} \right]$$

$$= \frac{1}{4} \left(\frac{3}{2} \right) + 0 + 0 + 0 = \frac{3}{8}$$

Q7 (2) - Single Correct

$$\sum_{n=3}^{\infty} \frac{1}{n^5 - 5n^3 + 4n} = \sum_{n=3}^{\infty} \frac{1}{n(n-1)(n-2)(n+1)(n+2)}$$

$$= \sum_{n=3}^{\infty} \frac{1}{4(n-2)(n-1)(n)(n+1)(n+2)}$$

$$= \sum_{n=3}^{\infty} \frac{1}{4} \left[\frac{1}{(n-2)(n-1)n(n+1)} - \frac{1}{(n-1)n(n+1)(n+2)} \right]$$

$$\text{Let } T_n = \frac{1}{4} \left[\frac{1}{(n-2)(n-1)n(n+1)} - \frac{1}{(n-1)n(n+1)(n+2)} \right]$$

Put $n = 1, 2, 3, \dots, n$

$$T_3 = \frac{1}{4} \left[\frac{1}{1 \cdot 2 \cdot 3 \cdot 4} - \frac{1}{2 \cdot 3 \cdot 4 \cdot 5} \right]$$

$$T_4 = \frac{1}{4} \left[\frac{1}{2 \cdot 3 \cdot 4 \cdot 5} - \frac{1}{3 \cdot 4 \cdot 5 \cdot 6} \right]$$

On adding all the above terms, we get

$$S_n = \frac{1}{4} \left[\frac{1}{1 \cdot 2 \cdot 3 \cdot 4} - \frac{1}{(n-1)n(n+1)(n+2)} \right]$$

$$= \lim_{n \rightarrow \infty} S_n = \frac{1}{4} \times \frac{1}{1 \cdot 2 \cdot 3 \cdot 4} = \frac{1}{96}$$

Q8 (3) - Single Correct

Area of all triangles

$$A = \frac{1}{2} \left[\frac{1}{3} \times 1 + \frac{1}{9} \times 2 + \frac{1}{27} \times 3 + \frac{1}{3^4} \times 4 + \dots \infty \right]$$

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$$= \frac{1}{2} \left[\frac{1}{3} + \frac{2}{3^2} + \frac{3}{3^3} + \frac{4}{3^4} + \dots \infty \right] = \frac{3}{8}$$

$$\Rightarrow 8A = 3$$

Q9 (2) - Single Correct

$H_1, H_2, H_3, \dots, H_{101}$ are in HP.

$\frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_{101}}$ are in AP. Let d be the common difference.

$$\Rightarrow \frac{1}{H_{i+1}} - \frac{1}{H_i} = d$$

$$\Rightarrow \frac{H_i - H_{i+1}}{H_i H_{i+1}} = d \dots (i)$$

$$\text{Now, } \sum_{i=1}^{100} (-1)^i \left(\frac{H_i + H_{i+1}}{H_i - H_{i+1}} \right)$$

$$= \sum_{i=1}^{100} (-1)^i \frac{(H_i + H_{i+1})}{H_i H_{i+1} \times d} \quad [\text{from Eq.(i)}]$$

$$= \sum_{i=1}^{100} \frac{(-1)^i}{d} \left(\frac{1}{H_{i+1}} + \frac{1}{H_i} \right)$$

$$= \frac{1}{d} \left[-\frac{1}{H_2} - \frac{1}{H_1} + \frac{1}{H_3} + \frac{1}{H_2} - \frac{1}{H_4} - \frac{1}{H_3} + \dots + \frac{1}{H_{101}} + \frac{1}{H_{100}} \right]$$

$$= \frac{1}{d} \left[\frac{1}{H_{101}} - \frac{1}{H_1} \right] = \frac{1}{d} [100d] = 100$$

Q10 (2) - Multiple Correct

$$\sum_{i=1}^k \frac{1}{d_i} = \frac{1}{n} \sum_{i=1}^k \left(\frac{n}{d_i} \right)$$

$$= \frac{1}{n} \{d_k + d_{k-1} + \dots + d_1\}$$

$$= \frac{1}{n} (72), \quad [d_1 + d_2 + \dots + d_k = 72, \text{ given}]$$

$$\therefore \sum_{i=1}^k \frac{1}{d_i} = \frac{72}{n} = \frac{72}{d_k}$$

Q11 (1,2,3,4) - Multiple Correct

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$$\therefore \alpha + \beta + \gamma = \frac{\pi}{2}$$

$$\Rightarrow \alpha + \beta = \frac{\pi}{2} - \gamma$$

$$\Rightarrow \cot(\alpha + \beta) = \cot\left(\frac{\pi}{2} - \gamma\right)$$

$$\Rightarrow \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta} = \tan \gamma$$

$$\Rightarrow \frac{\cot \alpha + \cot \beta}{\cot \alpha \cot \beta - 1} = \cot \gamma$$

$$\Rightarrow \cot \alpha + \cot \beta = \cot \alpha \cot \beta \cot \gamma - \cot \gamma$$

$$\Rightarrow \sum(\cot \alpha) = \Pi(\cot \alpha) \dots(i)$$

Also, $e^{\cot \alpha}, e^{\cot \beta}, e^{\cot \gamma}$ in GP

$$\therefore (e^{\cot \beta})^2 = e^{\cot \alpha} \cdot e^{\cot \gamma}$$

$$\therefore \log_e e^{2 \cot \beta} = \log_e e^{\cot \alpha} + \log_e e^{\cot \gamma}$$

$$2 \cot \beta = \cot \alpha + \cot \gamma$$

$$\text{As } \cot \alpha + \cot \beta + \cot \gamma = \cot \alpha \cot \beta \cot \gamma$$

$$\Rightarrow 2 \cot \beta + \cot \beta = \cot \alpha \cot \beta \cot \gamma \text{ [from Eq. (ii)]}$$

$$\Rightarrow 3 \cot \beta = \cot \alpha \cot \beta \cot \gamma$$

$$\Rightarrow \cot \alpha \cot \gamma = 3$$

$$\text{Also, } \Pi \cos \alpha = \frac{1}{2} \cos \gamma (2 \cos \alpha \cos \beta)$$

$$= \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)] \cos \gamma$$

$$= \frac{1}{4} \sin 2\gamma + \frac{1}{4} \left[\cos\left(\frac{\pi}{2} - 2\beta\right) + \cos\left(\frac{\pi}{2} - 2\alpha\right) \right]$$

$$\therefore \Pi \cos \alpha = \frac{1}{4} \Sigma(\sin 2\alpha)$$

Q12 (1) - Multiple Correct

$$S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}} = \sum_{n=1}^{99} \left(\frac{5^{100-2n}}{1+5^{100-2n}} \right)$$

$$= \frac{5^{98}}{1+5^{98}} + \frac{5^{96}}{1+5^{96}} + \frac{5^{94}}{1+5^{94}} + \dots + \frac{5^{-96}}{1+5^{-96}} + \frac{5^{-98}}{1+5^{-98}}$$

$$= \left(\frac{5^{98}}{1+5^{98}} + \frac{5^{-98}}{1+5^{-98}} \right) + \left(\frac{5^{96}}{1+5^{96}} + \frac{5^{-96}}{1+5^{-96}} \right) + \dots (49 \text{ th term}) + \frac{1}{2}$$

$$= 49 + \frac{1}{2} = 49.5$$

Q13 (1,2,3) - Multiple Correct

Given, If a, x, y, z, b are in AP.

$$xyz = 55$$

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And a, x, y, z, b are in HP

$$xyz = \frac{343}{55}$$

$\therefore a, x, y, z, b$ are in AP

Let b be the common difference of AP.

$$\therefore d = \frac{b-a}{4} \dots (i)$$

$$\Rightarrow x = a + \frac{b-a}{4} = \frac{3a+b}{4}$$

$$y = a + \frac{2(b-a)}{4} = \frac{2a+2b}{4} = \frac{a+b}{2}$$

$$\text{and } z = a + \frac{3(b-a)}{4} = \frac{a+3b}{4} \quad [\because xyz = 55, \text{ given}]$$

$$\frac{3a+b}{4} \times \frac{a+b}{2} \times \frac{a+3b}{4} = 55$$

$$\Rightarrow (3a+b)(a+b)(a+3b) = 55 \times 32 \dots (ii)$$

a, x, y, z, b are in HP $\Rightarrow \frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in AP.

Let D be the common difference.

$$\text{Then, } D = \frac{\frac{1}{b} - \frac{1}{a}}{4} \Rightarrow D = \frac{a-b}{4ab} \dots (iii)$$

$$\therefore \frac{1}{x} = \frac{1}{a} + \frac{a-b}{4ab} = \frac{4a+a-b}{4ab} = \frac{3b+a}{4ab}$$

$$x = \frac{4ab}{a+3b}$$

$$\frac{1}{y} = \frac{1}{a} + 2 \left(\frac{a-b}{4ab} \right) = \frac{1}{a} + \frac{a-b}{2ab}$$

$$= \frac{2b+a-b}{2ab} = \frac{a+b}{2ab}$$

$$\Rightarrow y = \frac{2ab}{a+b} \text{ and } \frac{1}{z} = \frac{1}{a} + 3 \left(\frac{a-b}{4ab} \right)$$

$$\frac{1}{z} = \frac{4b+3a-3b}{4ab} = \frac{3a+b}{4ab}$$

$$z = \frac{4ab}{3a+b}$$

$$\text{Now, } xyz = \frac{4ab}{a+3b} \times \frac{2ab}{a+b} \times \frac{4ab}{3a+b} = \frac{343}{55}$$

$$\Rightarrow \frac{32(ab)^3}{55 \times 32} = \frac{343}{55} \quad [\text{from Eq.}]$$

$$\Rightarrow (ab)^3 = 343 \Rightarrow ab = 7$$

$\therefore a$ and b are positive integers.

$\therefore$ Either $a = 1, b = 7$ or $a = 7, b = 1$

$$(1) a^2 + b^2 = 1^2 + 7^2 = 50$$

$$(2) \frac{b}{a} = \frac{7}{1} = 7$$

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$$(3) a + b = 1 + 7 = 8$$

Hence options (1), (2) and (3) are correct.

Q14 (2,3) - Multiple Correct

(1) $a, H_1, H_2, H_3, \dots, H_n, b$ in HP.

$\Rightarrow \frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \dots, \frac{1}{H_n}, \frac{1}{b}$ are in AP.

$$\Rightarrow \frac{1}{H_1} + \frac{1}{H_2} + \dots + \frac{1}{H_n} = n \left(\frac{\frac{1}{a} + \frac{1}{b}}{2} \right) = n \left(\frac{a+b}{2ab} \right) \neq n \cdot \left(\frac{2ab}{a+b} \right)$$

Hence, it is incorrect.

$$(2) 1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 = (1 + 2 + 3 + \dots + n)^2$$

Hence, it is correct

(3) $a, A_1, A_2, \dots, A_{2n}, b$ are in AP.

$$\Rightarrow A_1 + A_2 + A_3 + \dots + A_{2n} = \frac{2n(a+b)}{2} = n(a+b)$$

Hence, it is correct.

$$(4) g_1, g_2, \dots \infty; 4g_2 + 5g_3 = (4r + 5r^2) \text{ is minimum at } r = -\frac{2}{5},$$

given $g_1 = 1$

Hence, it is incorrect.

Q15 (1,2,3,4) - Multiple Correct

$a_1, a_2, a_3, \dots$ are in AP

$b_1, b_2, b_3, \dots$ are in AP

$$a_1 = 25, b_1 = 75 \text{ and } a_{100} + b_{100} = 100$$

(1) Let d and d_2 be common difference of two AP's. Hence, $a_1 + 99d_1 + b_1 + 99d_2 = 100$

$$\Rightarrow 99(d_1 + d_2) = 0 \Rightarrow d_1 = -d_2$$

Hence, statement (a) is correct.

$$(2) a_n + b_n = a_1 + (n-1)d_1 + b_1 + (n-1)d_2$$

$$= a_1 + b_1 + (n-1)(d_1 + d_2)$$

$$= (a_1 + b_1) = 100$$

Hence, statement (2) is correct.

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(3) $a_1 + b_1, a_2 + b_2, a_3 + b_3 + \dots$ are in AP is correct.

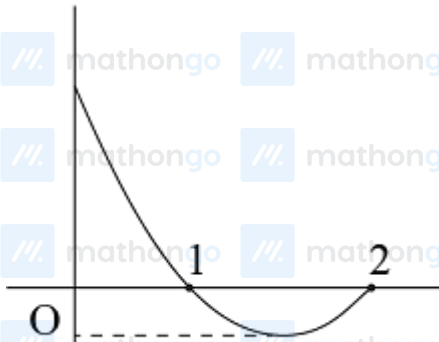
$$(4) \sum_{i=1}^{100} (a_f + b_r) = \sum_{i=1}^{100} 100 = 100 \times 100 = 10000$$

Hence, statement (4) is correct.

Q16 (1,2,3) - Multiple Correct

$$E = x^2 - 3x + 2 = (x-1)(x-2)$$

$$\therefore -\frac{D}{4a} \leq E \leq 2 \Rightarrow -\frac{1}{4} < E < 2$$



$$t_{10} = (3 + 9d)r^9 = [3 + 9(2)]\left(-\frac{1}{4}\right)^9$$

$$t_{10} = -\frac{(21)}{2^{18}}$$

$$\text{Also, } 4r + d - 1 = 4\left(-\frac{1}{4}\right) + 2 - 1 = 0$$

$$\Rightarrow 4r + d - 1 = 0$$

$$S = ab, (a+d)br, (a+2d)br^2, \dots$$

$$S = \frac{ab}{1-r} + \frac{dbr}{(1-r)^2}$$

$$\text{Here, } a = 3, b = 1, d = 2 \text{ and } r = -\frac{1}{4}$$

$$= \frac{3}{1+\frac{1}{4}} + \frac{2 \cdot 1 \cdot \left(-\frac{1}{4}\right)}{\left(1+\frac{1}{4}\right)^2} = \frac{12}{5} - \frac{8}{25} = \frac{60-8}{25} = \frac{52}{25}$$

Q17 (1,2,3,4) - Multiple Correct

$$\text{Given, } a + b + c = 8$$

$$\text{Clearly, } (a-b)^2 \geq 0$$

$$a^2 + b^2 - 2ab \geq 0$$

$$\Rightarrow a^2 + b^2 \geq 2ab$$

On adding $(a^2 + b^2)$ both sides, we get

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$$\Rightarrow 2a^2 + 2b^2 \geq a^2 + b^2 + 2ab$$

$$\Rightarrow (a^2 + b^2) \geq \frac{1}{2}(a + b)^2$$

$$\Rightarrow \frac{a^2 + b^2}{a + b} \geq \frac{1}{2}(a + b) \dots (i)$$

$$\text{Similarly, } \frac{b^2 + c^2}{b + c} \geq \frac{1}{2}(b + c) \dots (ii)$$

$$\frac{c^2 + a^2}{c + a} \geq \frac{1}{2}(c + a) \dots (iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$\Rightarrow \frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq \frac{1}{2}(2a + 2b + 2c)$$

$$\Rightarrow \frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq a + b + c$$

$$\Rightarrow \frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq 8$$

Hence, option (1) is correct.

and $\frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq 2$ is also true by result shown above.

Let $ab = x$, $bc = y$ and $ca = z$

$$\text{Since, } x^2 + y^2 + z^2 \geq xy + yz + zx$$

$$\Rightarrow a^2b^2 + b^2c^2 + c^2a^2 \geq (ab)(bc) + (bc)(ca) + (ca)(ab)$$

$$\Rightarrow a^2b^2 + b^2c^2 + c^2a^2 \geq abc(b + c + a)$$

$$\Rightarrow a^2b^2 + b^2c^2 + c^2a^2 \geq 8abc$$

$$\therefore \frac{a^m + b^m + c^m}{3} \geq \left(\frac{a + b + c}{3}\right)^m$$

$$\therefore \frac{a^2 + b^2 + c^2}{3} \geq \left(\frac{a + b + c}{3}\right)^2$$

$$\Rightarrow a^2 + b^2 + c^2 \geq \frac{3(8)^2}{9} \Rightarrow a^2 + b^2 + c^2 \geq \frac{64}{3}$$

$$\Rightarrow a^2 + b^2 + c^2 \geq 21\frac{1}{3}$$

Hence, $a^2 + b^2 + c^2 \geq 21\frac{1}{3}$

$$\therefore a^2 + b^2 + c^2 > 18$$

Q18 (2,3) - Multiple Correct

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Given, $\sin(x - y)$, $\sin x$ and $\sin(x + y)$ are in HP.

$$\begin{aligned} \Rightarrow \sin x &= \frac{2 \sin(x-y) \sin(x+y)}{\sin(x-y) + \sin(x+y)} \\ \Rightarrow \sin x &= \frac{2(\sin^2 x - \sin^2 y)}{2 \sin x \cos y} \\ \Rightarrow \sin^2 x \cos y &= \sin^2 x - \sin^2 y \\ \Rightarrow \sin^2 x (\cos y - 1) &= -\sin^2 y \\ \Rightarrow \sin^2 x &= \frac{\sin^2 y}{1 - \cos y} \\ \Rightarrow \sin^2 x &= \frac{\sin^2 y}{2 \sin^2 \frac{y}{2}} \left[\because 1 - \cos y = 2 \sin^2 \frac{y}{2} \right] \\ \Rightarrow \sin^2 x &= \frac{4 \sin^2(y/2) \cos^2(y/2)}{2 \sin^2 y/2} \left[\because \sin y = 2 \sin \frac{y}{2} \cos \frac{y}{2} \right] \\ \Rightarrow \sin^2 x &= 2 \cos^2 \frac{y}{2} \\ \Rightarrow \sin x &= \pm \sqrt{2} \cos \frac{y}{2} \\ \Rightarrow \frac{\sin x}{\cos \frac{y}{2}} &= \pm \sqrt{2} \\ \Rightarrow \sin x \sec \frac{y}{2} &= \pm \sqrt{2} \left[\because \frac{1}{\cos x} = \sec x \right] \end{aligned}$$

Hence, options (2) and (3) are correct.

Q19 (1,3,4) - Multiple Correct

Given, a , b and c are the first three terms of GP.

Let $a = a_1$, $b = a_1 r$, $c = a_1 r^2$

$\therefore$ HM of a and b is 12

$$12 = \frac{2ab}{a+b} = \frac{2a_1^2 r}{a_1(1+r)} = \frac{2a_1 r}{1+r}$$

and HM of b and c is 36.

$$\therefore 36 = \frac{2bc}{b+c} = \frac{2a_1 r^2}{(1+r)} \dots (i)$$

Dividing Eqs. (ii) by (i), we get

$$\frac{2a_1 r^2}{1+r} \times \frac{1+r}{2a_1 r} = \frac{36}{12} = 3 \dots (ii)$$

$$\Rightarrow r = 3$$

From Eq. (i), we have

$$\frac{2a_1 r}{1+r} = 12$$

$$\frac{6a_1}{4} = 12 \quad [\because r = 3]$$

$$a_1 = 8$$

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Now, option (1), $a_1 + r = 8 + 3 = 11$

$$\text{Option (2), } S_5 = \frac{8(3^5 - 1)}{3 - 1} \left[\because S_n = \frac{a(r^n - 1)}{r - 1}, r > 1 \right]$$

$$= 4(3^5 - 1) = 4 \times 242 = 968$$

$$a_1 = 8, d = 3$$

$$\text{Option (3), } T_8 = a_1 + 7d = 8 + 7 \times 3 = 8 + 21 = 29$$

Option (4), Let 648 is the n th term of GP.

$$\therefore T_n = ar^{n-1}$$

$$648 = 8 \times (3)^{n-1}$$

$$81 = 3^{n-1}$$

$$(3)^4 = 3^{n-1}$$

$$n - 1 = 4$$

$$n = 5$$

5th term of GP is 648

Hence options (1), (3) and (4) are correct

Q20 (1,4) - Multiple Correct

$$\text{Given, } \frac{a_2 a_3}{a_1 a_4} = \frac{a_2 + a_3}{a_1 + a_4} = 3 \left(\frac{a_2 - a_3}{a_1 - a_4} \right)$$

From first two equalities, we have

$$\frac{a_2 a_3}{a_1 a_4} = \frac{a_2 + a_3}{a_1 + a_4}$$

$$\Rightarrow \frac{a_2 + a_3}{a_2 a_3} = \frac{a_1 + a_4}{a_1 a_4}$$

$$\Rightarrow \frac{1}{a_3} + \frac{1}{a_2} = \frac{1}{a_4} + \frac{1}{a_1}$$

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_4} - \frac{1}{a_3} \dots (i)$$

Now, taking first and last equalities, we get

$$\frac{a_2 a_3}{a_1 a_4} = 3 \left(\frac{a_2 - a_3}{a_1 - a_4} \right)$$

$$\Rightarrow \frac{a_1 - a_4}{a_1 a_4} = 3 \left(\frac{a_2 - a_3}{a_2 a_3} \right)$$

$$\Rightarrow \frac{1}{a_4} - \frac{1}{a_1} = 3 \left(\frac{1}{a_3} - \frac{1}{a_2} \right)$$

$$\Rightarrow \frac{1}{a_4} - \frac{1}{a_3} + \frac{1}{a_3} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_1} = 3 \left(\frac{1}{a_3} - \frac{1}{a_2} \right)$$

$$\Rightarrow \left(\frac{1}{a_4} - \frac{1}{a_3} \right) + \left(\frac{1}{a_3} - \frac{1}{a_2} \right) + \left(\frac{1}{a_2} - \frac{1}{a_1} \right) = 3 \left(\frac{1}{a_3} - \frac{1}{a_2} \right)$$

$$\Rightarrow \left(\frac{1}{a_2} - \frac{1}{a_1} \right) + \left(\frac{1}{a_3} - \frac{1}{a_2} \right) + \left(\frac{1}{a_2} - \frac{1}{a_1} \right) = 3 \left(\frac{1}{a_3} - \frac{1}{a_2} \right) \text{ [from Eq. (i)]}$$

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$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2}$$

From Eqs. (i) and (ii), we get

$$\frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} = \frac{1}{a_4} - \frac{1}{a_3}$$

$$\Rightarrow \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \frac{1}{a_4} \text{ are in AP. [option (1)]}$$

$$\Rightarrow a_1, a_2, a_3, a_4 \text{ are in HP. [option (4)]}$$

Hence, options (1) and (4) are correct.

Q21 (2,3) - Multiple Correct

Given,

$$x^2 + 9y^2 + 25z^2 = xyz \left(\frac{15}{x} + \frac{5}{y} + \frac{3}{z} \right)$$

$$\Rightarrow x^2 + 9y^2 + 25z^2 = 15yz + 5xz + 3xy$$

$$\Rightarrow x^2 + (3y)^2 + (5z)^2 - 3xy - 15yz - 5xz = 0$$

$$\Rightarrow \frac{1}{2} [2x^2 + 2 \cdot (3y)^2 + 2(5z)^2 - 2(3xy) - 2(15yz) - 2(5xz)] = 0$$

$$\Rightarrow \frac{1}{2} [\{x^2 + (3y)^2 - 2(3xy)\} + \{x^2 + (5z)^2 - 2(5xz)\} + \{(3y)^2 + (5z)^2 - 2(15yz)\}] = 0$$

$$\Rightarrow \frac{1}{2} [(x - 3y)^2 + (x - 5z)^2 + (3y - 5z)^2] = 0$$

$$\Rightarrow x - 3y = 0, x - 5z = 0, 3y - 5z = 0$$

$$\Rightarrow x = 3y, x = 5z, 3y = 5z$$

$$\Rightarrow x : y : z = 15 : 5 : 3 \text{ [option (C)]}$$

$$\Rightarrow \frac{1}{x} : \frac{1}{y} : \frac{1}{z} = 1 : 3 : 5$$

$$\Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in AP}$$

$$\Rightarrow x, y \text{ and } z \text{ are in HP.}$$

Q22 (2,4) - Multiple Correct

Given, $a_1 = 2$

$$\text{and } a_n = \frac{a_{n-1}^2}{a_{n-2}} \Rightarrow \frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}} = r \text{ (say)}$$

Since, $a_1, a_2, a_3, a_4, a_5, \dots$ are in GP.

Let $a_2 = y$, then for $n = 3$, we have

$$\frac{a_3}{a_2} = \frac{a_2}{a_1} \Rightarrow a_3 = \frac{y^2}{2}$$

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So, the series is $2, y, \frac{y^2}{2}, \frac{y^3}{4}, \frac{y^4}{8}$.

Hence, common ratio $(r) = \frac{y}{2}$

It is given that, $a_5 \leq 162$

$$\Rightarrow ar^4 \leq 162$$

$$\Rightarrow 2 \cdot \left(\frac{y}{2}\right)^4 \leq 162 \Rightarrow \frac{y^4}{8} \leq 162$$

$$\Rightarrow y^4 \leq 1296 \Rightarrow y \leq 6$$

Also, a_2 and a_5 are positive integers i.e. y and $\frac{y^4}{8}$ are positive

integers Since, y must be even, then only $\frac{y^4}{8}$ will be an integer.

So, the possible values of y are 4 and 6.

$[\because y \neq 2$ as terms are distinct]

Hence, the possible value of $a_5 = \frac{y^4}{8}$ are $\frac{4^4}{8}, \frac{6^4}{8}$ i.e. 32, 162.

Q23 (1,3,4) - Multiple Correct

Given equation is $x^3 + px^2 + qx - 1 = 0$

Let α, β and γ be the roots of the equation.

$$\therefore \alpha + \beta + \gamma = -p \dots (i)$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q \dots (ii)$$

$$\text{and } \alpha\beta\gamma = 1 \dots (iii)$$

Since, α, β, γ form an increasing GP.

$$\therefore \alpha < \beta < \gamma$$

$$\text{Let } \alpha = a_1\beta = ar_1\gamma = ar^2$$

$$\text{From Eq. (i), } a(1 + r + r^2) = -p \dots (iv)$$

$$\text{From Eq. (ii), } a^2r + a^2r^3 + a^2r^2 = q$$

$$a^2r(1 + r + r^2) = q \dots (v)$$

$$\text{From Eq. (iii), } a^3r^3 = 1$$

$$\Rightarrow (ar)^3 = 1$$

$$\Rightarrow ar = 1$$

$$\text{i.e. } \beta = 1$$

Thus, the one root is unity.

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Hence, option (3) is correct.

On dividing Eq (v) by Eq. (iv), we get

$$\frac{a^2r(1+r+r^2)}{a(1+r+r^2)} = \frac{q}{-p}$$

$$\Rightarrow ar = \frac{q}{-p} \Rightarrow 1 = \frac{q}{-p} [\because ar = 1]$$

$$\Rightarrow p + q = 0$$

Hence, option (1) is correct.

$$\alpha < \beta < \gamma$$

$$\Rightarrow \alpha < 1 < \gamma$$

So, one root is less than 1 and one root is greater than 1.

Hence, option (4) is correct.

Q24 (2,4) - Multiple Correct

Given, $\log a, \log b, \log c$ are in AP

$$\therefore 2 \log b = \log a + \log c$$

$$b^2 = ac \dots (i)$$

So, a, b, C are in GP.

and $(\log a - \log 2b), (\log 2b - \log 3c), (\log 3c - \log a)$

are in AP.

$$\therefore 2(\log 2b - \log 3c) = \log a - \log 2b + \log 3c - \log a$$

$$\Rightarrow 2 \log \frac{2b}{3c} = \log \frac{3c}{2b}$$

$$\Rightarrow \left(\frac{2b}{3c}\right)^2 = \frac{3c}{2b}$$

$$\Rightarrow (2b)^3 = (3c)^3 \Rightarrow 2b = 3c$$

$$\Rightarrow \frac{b}{c} = \frac{3}{2}$$

$$\Rightarrow \frac{a}{b} = \frac{3}{2} \text{ [from Eq. (i)]}$$

$$a : b : c = 9 : 6 : 4$$

So, a, b, c cannot be length of the sides of a triangle.

Lol terms $a_1, a_2, a_3, \dots, a_{2n+1}$ are in AP and $a_{2n+1} \dots a_{4n+1}$ are in GP. Common difference, $d = 2$, Common

$$\text{ratio, } r = \frac{1}{2}$$

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Q25 (2) - Paragraph 1

Given middle term of AP = Middle term of GP

i.e. $a_{n+1} = a_{3n+1}$

$$a_1 + 2n = a_{2n+1} \times \left(\frac{1}{2^n}\right)$$

$$\Rightarrow a_1 2^n + 2 \cdot n \cdot 2^n = a_1 + 4n$$

$$\Rightarrow a_1 (2^n - 1) = 4n - 2n \cdot 2^n$$

$$\Rightarrow a_1 = \left(\frac{4n - 2n \cdot 2^n}{2^n - 1}\right)$$

Q26 (1) - Paragraph 1

Now, $a_{2n+1} = a_1 + 4n$

$$= \left(\frac{4n - 2n \cdot 2^n}{2^n - 1} + 4n\right) = \frac{2n \cdot 2^n}{2^n - 1} = \frac{n \cdot 2^{n+1}}{2^n - 1}$$

Q27 (2) - Paragraph 2

$$g(n) - f(n) = \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2} \left[\frac{2n+1}{3} - 1\right] = \frac{n(n+1)(2n-2)}{2 \times 3} = \frac{n(n^2-1)}{3}$$

which is divisible by n i.e. $\frac{n^2-1}{3}$ is integer. For $n = 1, 0$,

$$n = 2, 1; \quad n = 3, \frac{8}{3}; \quad n = 4, 5; \quad n = 5, 8$$

$$n = 6, \frac{35}{3} \text{ not an integer.}$$

Hence, for $n = 4$, is greatest even natural number which divides $g(n) - f(n)$

Q28 (4) - Paragraph 2

$$f(n) + 3g(n) + h(n)$$

$$= \frac{n(n+1)}{2} + \frac{3n(n+1)(2n+1)}{6} + \left[\frac{n(n+1)}{2}\right]^2$$

$$= \frac{n(n+1)}{2} \left[1 + (2n+1) \frac{n(n+1)}{2}\right] = \frac{n(n+1)}{2} \left[\frac{4+4n+n^2+n}{2}\right]$$

$$= \frac{n(n+1)}{2} \left[\frac{(n+1)(n+4)}{2}\right]$$

So, $\frac{(n+1)(n+4)}{2}$ is integer, Now if n is odd, then $n+1$ is even. If n is even, then $n+4$ is even.

Hence, $f(n) + 3g(n) + h(n)$ is divisible by $f(n)$ for all $n \in N$

Q29 (2) - Paragraph 3

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Hence, minimum possible value of a is $\frac{5}{26}$

Q30 (4) - Paragraph 3

Let $S = \sum_{r=5}^{\infty} \frac{1}{t_r}$

Also, $t_r = (r-1)(r-2)(r-3)(r-4)$

Let, $S_n = \sum_{r=5}^n \frac{1}{(r-1)(r-2)(r-3)(r-4)}$

$$S_n = \frac{1}{3} \left[\frac{(4-1)}{1 \cdot 2 \cdot 3 \cdot 4} + \frac{(5-2)}{2 \cdot 3 \cdot 4 \cdot 5} + \dots + \frac{(n-1)-(n-4)}{(n-4)(n-3)(n-2)} \right]$$

$$= \frac{1}{3} \left[\left(\frac{1}{1 \cdot 2 \cdot 3} - \frac{1}{2 \cdot 3 \cdot 4} \right) + \left(\frac{1}{2 \cdot 3 \cdot 4} - \frac{1}{3 \cdot 4 \cdot 5} \right) + \dots \right]$$

$$S_n = \frac{1}{3} \left[\frac{1}{1 \cdot 2 \cdot 3} - \frac{1}{(n-1)(n-2)(n-3)} \right]$$

$$S = \lim_{n \rightarrow \infty} \delta_n = \frac{1}{3} \left[\frac{1}{6} - \frac{1}{\infty} \right]$$

$$S = \frac{1}{18}$$

Q31 (3) - Paragraph 4

Let α, β, γ be the roots of the equation

$$2x^3 - 19x^2 + 57x + k$$

$$\alpha + \beta + \gamma = \frac{19}{2} \quad \dots (i)$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{57}{2} \quad \dots (ii)$$

$$\alpha\beta\gamma = -\frac{k}{2} \quad \dots (iii)$$

Let $\alpha = a, \beta = ar, \gamma = ar^2$

From Eq. (i),

$$a + ar + ar^2 = \frac{19}{2}$$

$$\Rightarrow a(1 + r + r^2) = \frac{19}{2} \quad \dots (iv)$$

From Eq. (ii),

$$a \cdot ar + ar \cdot ar^2 + ar^2 \cdot a = \frac{57}{2}$$

$$\Rightarrow a^2r + a^2r^3 + a^2r^2 = \frac{57}{2}$$

$$\Rightarrow a^2(r + r^2 + r^3) = \frac{57}{2}$$

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$$\Rightarrow a^2 r (1 + r + r^2) = \frac{57}{2} \dots (v)$$

$$\text{From Eq. (iii), } a^3 r^3 = -\frac{k}{2}$$

$$\Rightarrow (ar)^3 = -\frac{k}{2}$$

On dividing Eq. (v) by Eq. (iv), we have

$$ar = \frac{57}{2} \times \frac{2}{19} = \frac{57}{19} = 3$$

$$\Rightarrow ar = 3 \dots (vi)$$

On dividing Eq. (v) by Eq. (iv), we have

$$ar = \frac{57}{2} \times \frac{2}{19} = \frac{57}{19} = 3$$

$$\Rightarrow ar = 3$$

From Eqs. (vi) and (vii),

$$(3)^3 = -\frac{k}{2} \Rightarrow 27 = -\frac{k}{2}$$

$$\therefore k = -54$$

Q32 (4) - Paragraph 4

If GP are increasing, then

$$a < ar < ar^2$$

$$\text{From Eq. (iv), } a + ar + ar^2 = \frac{19}{2}$$

$$\Rightarrow a + 3 + 3r = \frac{19}{2}$$

$$\Rightarrow a + 3r = \frac{19}{2} - 3 = \frac{13}{2} \dots (viii)$$

$$\text{From Eqs. (vii) and (viii), } a + 3r = \frac{13}{2}$$

$$\Rightarrow \frac{3}{r} + 3r = \frac{13}{2}$$

$$\Rightarrow 6r^2 - 13r + 6 = 0 \quad [\because ar = 3]$$

$$\Rightarrow r = \frac{13 \pm \sqrt{169 - 144}}{12}$$

$$\Rightarrow r = \frac{13 \pm 5}{12} = \frac{18}{12}, \frac{8}{12} \Rightarrow r = \frac{3}{2}, \frac{2}{3}$$

Since, GP is increasing

$$\therefore r = \frac{3}{2} > 1, a = 2$$

$$\therefore \text{Sum of } n \text{ terms} = \frac{2 \left[\left(\frac{3}{2} \right)^n - 1 \right]}{\frac{1}{2}} = 4 \left[\left(\frac{3}{2} \right)^n - 1 \right]$$

Q33 (2) - Paragraph 5

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Explanation (For Question 34, 35)Given, $T_1 = 0, T_2 = 1, T_3 = 1$ and $T_4 = 2$

$$\therefore T_n = A \cdot \alpha^{n-1} + B \cdot \beta^{n-1} \dots (i)$$

Put $n = 1$ in Eq. (i), we get

$$0 = A + B \text{ i.e. } A = -B \dots (ii)$$

Put $n = 2$ in Eq. (i), we get

$$1 = A\alpha + B\beta$$

Put $n = 3$ in Eq. (i), we get

$$1 = A\alpha^2 + B\beta^2$$

Put $n = 4$ in Eq. (i), we get

$$2 = A\alpha^3 + B\beta^3$$

From Eqs. (ii) and (iii),

$$\frac{1}{A} = (\alpha - \beta)$$

From Eqs. (ii) and (iv),

$$\frac{1}{A} = (\alpha^2 - \beta^2)$$

$$\Rightarrow \frac{1}{A} = \frac{1}{A}(\alpha + \beta) \text{ [from Eq. (vi)]}$$

$$\Rightarrow \alpha + \beta = 1$$

$$\text{Now, } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

From Eqs. (ii) and (v), we get

$$\frac{2}{A} = (\alpha^3 - \beta^3)$$

$$\Rightarrow 2 = (\alpha^2 + \beta^2 + \alpha\beta)$$

So, option (2) is correct.

Q34 (3) - Paragraph 5The quadratic equation having roots α and β , we have

$$\alpha + \beta = 1, \alpha\beta = -1$$

From the equation $x^2 - (\alpha + \beta)x + \alpha\beta$, we get

$$x^2 - (1)x - 1 = x^2 - x - 1$$

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Q35 (1) - Matrix Match

$$(i) p^2 + 9q^2 + 25r^2 = pqr \left(\frac{15}{p} + \frac{5}{q} + \frac{3}{r} \right) = 15qr + 5pr + 3pq$$

$$\Rightarrow 2p^2 + 18q^2 + 50r^2 - 30qr - 10pr - 6pq = 0$$

$$\Rightarrow (p - 3q)^2 + (p - 5r)^2 + (3q - 5r)^2 = 0$$

$$\Rightarrow p = 3q$$

$$p = 5r$$

$$4q = 5r$$

$$\Rightarrow p = 3q = 5r = \lambda \text{ (say)}$$

$$\text{Now, } \frac{1}{p} + \frac{1}{r} = \frac{1}{\lambda} + \frac{5}{\lambda} = \frac{6}{\lambda} = 2 \times \frac{1}{q}$$

Hence, p, q and r are in HP.

$$(ii) 2 \log(r - p) = \log(p + r) + \log(p - 2q + r)$$

$$(r - p)^2 = (p + r)(p - 2q + r) \Rightarrow 2pq + 2qr = 4pr$$

$$\Rightarrow \frac{1}{r} + \frac{1}{p} = \frac{2}{q}$$

Hence, p, q and r are in HP.

$$(iii) f(x) = px^2 + qx + r$$

$$\Rightarrow f'(x) = 2px + q$$

$$\therefore f(1) = f(-1)$$

$$\Rightarrow q = 0$$

$$f'(p) = 2p^2 + q = 2p^2 \text{ as } q = 0$$

$$f'(q) = 2pq + q = 0$$

$$f'(r) = 2pr + q = 2pr$$

As, $f'(p), f'(q), f'(r)$ are in AP i.e. $2p^2, 0, 2pr$ are in AP

$p, 0, r$ are in AP

p, q, r are in AP

(iv) $p, q, r \in \mathbb{R}^+$ such that

$$p + q + r = 25 \dots(i)$$

$$2 < p, q, r < 18 \dots(ii)$$

$2, p, q$ are in AP.

$$\Rightarrow 2p = 2 + q \dots(iii)$$

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$q, r, 18$ are in GP.

$$\Rightarrow r^2 = 18q \dots (iv)$$

from Eqs. (i), (iii), (iv), we get

$$1 + \frac{r^2}{36} + \frac{r^2}{18} + r = 25$$

$$r^2 + 2r^2 + 36r = 24 \times 36$$

$$r^2 + 12r - 288 = 0$$

$$(r + 24)(r - 12) = 0$$

$\Rightarrow r = 12$ excluding $r = -24$ as r is positive.

$$\Rightarrow q = 8, p = 5$$

Hence, $p \neq q \neq r$

(i) $\rightarrow t$, (ii) $\rightarrow t$, (iii) $\rightarrow r_1$ (iv) $\rightarrow q$

Q36 (1) - Integer Type

$$2acx^2 + 2abcx + (a + c) = 0 \dots (i)$$

If α, β are roots of Eq. (i). Then,

$$\alpha + \beta = -b$$

$$\alpha\beta = \frac{a+c}{2ac}$$

$\frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in AP. Therefore, we have

$$2b = \frac{a+b}{1-ab} + \frac{b+c}{1-bc}$$

$$2b(1-ab)(1-bc) = a+b-abc-b^2c+b+c-ab^2-abc$$

$$\Rightarrow 2b - 2ab^2 - 2b^2c + 2ab^3c = a + 2b + c - b^2c - ab^2 - 2abc$$

$$\Rightarrow 2abc(1+b^2) = (a+c)(1-b^2+2b^2)$$

$$\Rightarrow 2abc = a + c$$

$$b = \frac{a+c}{2ac} \dots (ii)$$

Now,

$$(1+\alpha)(1+\beta) = 1 + \alpha + \beta + \alpha\beta$$

$$= 1 - b + \frac{a+c}{2ac} = 1 - b + b \quad [\text{from Eq.}]$$

$$= 1$$

Q37 (5) - Integer Type

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$$\frac{n(n+1)}{2} - k - (k+1) = 1224$$

$$n(n+1) - 4k = 2450$$

$$4k = n(n+1) - 2450$$

By analysis, $n = 50$

i.e. $4k = 2550 - 2450$

$$4k = 100$$

$$k = 25$$

Then, $k - 20 = 5$

Q38 (6) - Integer Type

$$S_n = (1 + 2T_n)(1 - T_n)$$

$$n = 1, S_1 = T_1 = (1 + 2T_1)(1 - T_1)$$

$$\Rightarrow T_1^2 = \frac{1}{2}$$

$$\Rightarrow T_1 = \frac{1}{\sqrt{2}} [\text{as } T_i > 0]$$

$$\text{If } n = 2, S_2 = (1 + 2T_2)(1 - T_2)$$

$$T_1 + T_2 = 1 + T_2 - 2T_2^2$$

$$T_2^2 = \frac{1}{2} - \frac{1}{2\sqrt{2}}$$

$$T_2^2 = \frac{\sqrt{2}-1}{2\sqrt{2}} = \frac{2-\sqrt{2}}{4} = \frac{\sqrt{4}-\sqrt{2}}{4} = \frac{\sqrt{a}-\sqrt{b}}{4}$$

$$\Rightarrow a = 4, b = 2$$

Then, $a + b = 6$

Q39 (9) - Integer Type

Let a be first term, d be common difference, then

$$T_{20} = a + 19d = \log_{10} 20 \dots (i)$$

$$T_{32} = a + 31d = \log_{10} 32 \dots (ii)$$

$$d = \frac{\log_{10} 2}{3} - \frac{1}{12}$$

$$\Rightarrow a = -\frac{16}{3} \log_{10} 2 + \frac{31}{12}$$

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Let T_{n+1} be rational i.e.

$$a + nd = \left(\frac{n-16}{3}\right) \log_{10} 2 + \frac{31-n}{12} \text{ is rational}$$

$$\text{If } n = 16 \text{ i.e. } T_{17} = \frac{15}{12} = \frac{5}{4} = \frac{p}{q}$$

$$\therefore p + q = 5 + 4 = 9$$

Q40 (3) - Integer Type

Given, a, b and c are positive integers

$$(a-1)^2 \geq 0 \Rightarrow a^2 + 1 \geq 2a$$

$$\Rightarrow \frac{a^2+1}{b+c} \geq \frac{2a}{b+c} \dots(i)$$

$$(b-1)^2 \geq 0 \Rightarrow b^2 + 1 \geq 2b$$

$$\Rightarrow \frac{b^2+1}{a+c} \geq \frac{2b}{a+c} \dots(ii)$$

$$\text{Also, } (c-1)^2 \geq 0$$

$$\Rightarrow c^2 + 1 \geq 2c \dots(iii)$$

$$\Rightarrow \frac{c^2+1}{a+b} \geq \frac{2c}{a+b}$$

On adding Eqs. (i), (ii) and (iii), we get

$$\frac{a^2+1}{b+c} + \frac{b^2+1}{a+c} + \frac{c^2+1}{a+b} \geq \frac{2a}{b+c} + \frac{2b}{a+c} + \frac{2c}{a+b}$$

$$\Rightarrow \frac{a^2+1}{b+c} + \frac{b^2+1}{a+c} + \frac{c^2+1}{a+b} \geq \frac{2a}{b+c} + \frac{2b}{a+c} + \frac{2c}{a+b}$$

$$\Rightarrow \frac{a^2+1}{b+c} + \frac{b^2+1}{a+c} + \frac{c^2+1}{a+b} + 6$$

$$\geq \left(\frac{2a}{b+c} + 2\right) + \left(\frac{2b}{a+c} + 2\right) + \left(\frac{2c}{a+b} + 2\right)$$

$$\Rightarrow \frac{a^2+1}{b+c} + \frac{b^2+1}{a+c} + \frac{c^2+1}{a+b} + 6$$

$$\geq (2a + 2b + 2c) \left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b}\right)$$

$$\Rightarrow \frac{a^2+1}{b+c} + \frac{b^2+1}{a+c} + \frac{c^2+1}{a+b} + 6$$

Hints & Solutions

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$$\begin{aligned} &\geq [(b+c) + (a+c) + (a+b)] \left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b} \right) \\ &\Rightarrow \frac{a^2+1}{b+c} + \frac{b^2+1}{a+c} + \frac{c^2+1}{a+b} + 6 \geq (3)^2 \\ &\left[\because \left(a_1 + a_2 + \dots + a_n \right) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \geq n^2 \right] \\ &\Rightarrow \frac{a^2+1}{b+c} + \frac{b^2+1}{c+a} + \frac{c^2+1}{a+b} \geq 3 \end{aligned}$$

Hence, minimum value is 3.

Q41 (8) - Integer Type

Given numbers are $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and $a^{10}; a > 0$

We know that, $AM \geq GM \Rightarrow \frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10}}{8}$

$$\begin{aligned} &\geq (a^{-5} \cdot a^{-4} \cdot a^{-3} \cdot a^{-3} \cdot a^{-3} \cdot 1 \cdot a^8 \cdot a^{10})^{1/8} \\ &\Rightarrow \frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10}}{8} \geq (a^{-18} \cdot a^{18})^{1/8} \\ &\Rightarrow a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10} \geq 8 \end{aligned}$$

Hence, minimum value of the sum of given numbers is 8.

Q42 (3) - Integer Type

Since, S_K denote the sum of the infinite geometric series whose first term is $\frac{(k-1)}{k!}$ and common ratio is $\frac{1}{k}$.

$$\begin{aligned} \therefore S_K &= \frac{\frac{k-1}{k!}}{1 - \frac{1}{k}} \left[\because S_\infty = \frac{a}{1-r} \right] \\ S_k &= \frac{k-1}{k!} \times \frac{k}{k-1} \Rightarrow S_k = \frac{1}{(k-1)!} \end{aligned}$$

Now

$$\begin{aligned} \sum_{k=2}^{100} |(k^2 - 3k + 1) S_k| &= \sum_{k=2}^{100} \left| (k^2 - 3k + 1) \times \frac{1}{(k-1)!} \right| \left[\because S_k = \frac{1}{(k-1)!} \right] \\ &= \sum_{k=2}^{100} \left| \frac{(k^2 - 2k + 1) - k}{(k-1)!} \right| = \sum_{k=2}^{100} \left| \frac{(k-1)^2 - k}{(k-1)!} \right| \\ &= \sum_{k=2}^{100} \left| \frac{(k-1)^2}{(k-1)!} - \frac{k}{(k-1)!} \right| = \sum_{k=2}^{100} \left| \frac{k-1}{(k-2)!} - \frac{k}{(k-1)!} \right| \\ &= \sum_{k=2}^{100} \left| \frac{(k-2)+1}{(k-2)!} - \frac{(k-1)+1}{(k-1)!} \right| \\ &= \sum_{k=2}^{100} \left| \frac{k-2}{(k-2)!} + \frac{1}{(k-2)!} - \frac{k-1}{(k-1)!} - \frac{1}{(k-1)!} \right| \\ &= \sum_{k=2}^{100} \left| \frac{k-2}{(k-2)!} + \frac{1}{(k-2)!} - \frac{1}{(k-2)!} - \frac{1}{(k-1)!} \right| \\ &= \sum_{k=3}^{100} \left[\frac{1}{(k-3)!} - \frac{1}{(k-2)!} \right] + \sum_{k=2}^{100} \left[\frac{1}{(k-2)!} - \frac{1}{(k-1)!} \right] \end{aligned}$$

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Hints & Solutions

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$$\begin{aligned}
 &= \left[\left\{ \left(1 - \frac{1}{1!}\right) + \left(\frac{1}{1!} - \frac{1}{2!}\right) + \left(\frac{1}{2!} - \frac{1}{3!}\right) + \cdots + \left(\frac{1}{97!} - \frac{1}{98!}\right) \right\} \right. \\
 &\quad \left. + \left\{ \left(1 - \frac{1}{1!}\right) + \left(\frac{1}{1!} - \frac{1}{2!}\right) + \left(\frac{1}{2!} - \frac{1}{3!}\right) + \cdots + \left(\frac{1}{98!} - \frac{1}{99!}\right) \right\} \right] \\
 &= \left[\left(1 - \frac{1}{98!}\right) + \left(1 - \frac{1}{99!}\right) \right] = \left[2 - \frac{1}{98!} - \frac{1}{99!} \right] \\
 &= \left[2 - \frac{1}{98!} \left(1 + \frac{1}{99}\right) \right] = \left[2 - \frac{1}{98!} \times \frac{100}{99} \right] \\
 &\sum_{k=2}^{100} |(k^2 - 3k + 1) S_k| = \left[2 - \frac{100}{99!} \right] \dots (i) \\
 &[\because (n+1) \times n! = n+1!] \\
 &\text{Now, } \frac{100^2}{100!} + \sum_{k=1}^{100} |(k^2 - 3k + 1) S_k| = \frac{100}{99!} + |(1^2 - 3 \times 1 + 1) S_1| + \sum_{k=2}^{100} |(k^2 - 3k + 1) S_k| \\
 &= \frac{100}{99!} + |(-1) \cdot S_1| + \sum_{k=2}^{100} |(k^2 - 3k + 1) S_k| \quad [\text{from Eq. (i)}] \\
 &= \frac{100}{99!} + S_1 + 2 - \frac{100}{99!} \\
 &= S_1 + 2 \because S_k = \frac{1}{k-1!} S_1 = \frac{1}{1-1!} = \frac{1}{0!} = 1 \Big] \\
 &= 1 + 2 = 3
 \end{aligned}$$

Q43 (9) - Integer Type

Given, a, x , b are in AP

$$\Rightarrow 2x = a + b \dots (i)$$

a, y , b are in GP

$$\Rightarrow y^2 = ab \dots (ii)$$

and a , z , b are in HP.

$$\Rightarrow z = \frac{2ab}{a+b} \dots (iii)$$

$$\Rightarrow z = \frac{2y^2}{2x} \quad [\text{from Eqs. (i) and (ii)}]$$

$$\Rightarrow z = \frac{y^2}{x}$$

$$\Rightarrow y^2 = xz \dots (iv)$$

Since, we have $x = 9z \dots (v)$

From Eqs. (iv) and (v), we get

$$9y^2 = x^2$$

$$\therefore \frac{x^2}{y^2} = 9$$

Q44 (0) - Integer Type

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Hints & Solutions

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Given, $\frac{a^n+b^n}{a^{n-1}+b^{n-1}}$ is the HM between a and b.

$$\therefore \frac{a^n+b^n}{a^{n-1}+b^{n-1}} = \frac{2ab}{a+b}$$

$$\Rightarrow (a+b)(a^n+b^n) = 2ab(a^{n-1}+b^{n-1})$$

$$\Rightarrow a^{n+1}+ab^n+ba^n+b^{n+1} = 2ab(a^{n-1}+b^{n-1})$$

$$\Rightarrow a^{n+1}+ab^n+ba^n+b^{n+1} = 2a^nb+2b^na$$

$$\Rightarrow a^{n+1}+b^{n+1} = a^nb+b^na$$

$$\Rightarrow (a^{n+1}-a^nb) + (b^{n+1}-b^na) = 0$$

$$\Rightarrow a^n(a-b) + b^n(b-a) = 0$$

$$\Rightarrow a^n(a-b) - b^n(a-b) = 0$$

$$\Rightarrow (a-b)(a^n-b^n) = 0$$

$$\Rightarrow a^n-b^n = 0 \quad [\because a-b \neq 0]$$

$$\Rightarrow a^n = b^n$$

$$\Rightarrow \left(\frac{a}{b}\right)^n = 1 = \left(\frac{a}{b}\right)^0$$

$$\therefore n = 0$$